

This unit is assessed by means of a written examination paper of 1 hour 35 minutes duration. The paper will consist of objective, short-answer and long-answer questions. Students may be required to apply their knowledge and understanding of physics to situations that they have not encountered before. The total number of marks available for this examination paper is 80. It contributes 40% to IA2 and 20% to the IAL in Physics.

It is recommended that students have access to a scientific calculator for this paper.

Students will be provided with the formulae sheet shown in *Appendix 5*. Any other physics formulae that are required will be stated in the question paper.

The quality of written communication will be assessed in the context of this unit through questions which are labelled with an asterisk (*). Students should take particular care with spelling, punctuation and grammar, as well as the clarity of expression, on these questions.

7.3 Transport on Track (TRA)

A study of a modern rail transport system with an emphasis on safety and control:

- track circuits and signalling
- sensing speed
- mechanical braking
- regenerative and eddy current braking
- crash-proofing.

Students will use mathematical models to describe the behaviour of moving vehicles and to model electromagnetic induction and capacitor discharge.

There are opportunities for students to develop information and communication technology skills.

There are opportunities for students to discuss ethical, environmental and other issues relating to decisions about transport taken by government, by transport companies and by individuals.

Students will be assessed on their ability to:		Suggested experiments
73	use the expression $p = mv$	
74	investigate and apply the principle of conservation of linear momentum to problems in one dimension	Use of, for example, light gates and air track to investigate momentum
75	investigate and relate net force to rate of change of momentum in situations where mass is constant (Newton's second law of motion)	Use of, for example, light gates and air track to investigate change in momentum
78	explain and apply the principle of conservation of energy, and determine whether a collision is elastic or inelastic	
87	investigate and use the expression $C = Q/V$	Use a Coulometer to measure charge stored
89	investigate and recall that the growth and decay curves for resistor–capacitor circuits are exponential, and know the significance of the time constant RC	
90	recognise and use the expression $Q = Q_0 e^{-t/RC}$ and derive and use related expressions, for exponential discharge in RC circuits, for example, $I = I_0 e^{-t/RC}$	Use of data logger to obtain I – t graph